

BCAN 14203

Following Paper ID and Roll No. to be filled in your Answer Book.

**Roll
No.**

[illegible]

(Even Semester)

[Maximum Marks : 60]

SECTION-A

1. Attempt all parts of the following : $8 \times 1 = 8$
- (a) Define algorithm.
 - (b) What is the time complexity of insertion sort?
 - (c) What are the applications of Greedy method?
 - (d) Define activity selection problem.
 - (e) Define Hamiltonian circuit problem.
 - (f) What is dynamic programming?

[P. T. O.]

- (g) Explain live node and dead node.
- (h) Define single source shortest path.

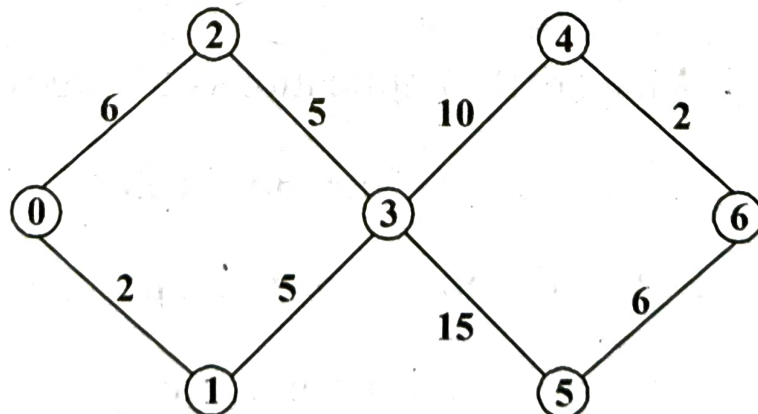
SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Perform selection sort on the following set of elements :

50, 55, 20, 45, 17, 25, 28, 30

- (b) Write down the algorithm of finding maximum and minimum element with the help of divide and conquer. Explain it with the help of an example.
- (c) Explain N-queen problem.
- (d) Find the single source shortest path using Dijkstra algorithm :



SECTION-C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Perform heap sort on the following set of elements :

6, 10, 5, 12, 3, 9, 20, 15, 8, 18

- (b) Explain the types of asymptotic notation in detail.

- (c) What is bubble sort? Explain the time complexity of bubble sort in detail.

4. (a) Find out the minimum cost and path using travelling salesman problem :

	A	B	C	D
A	0	4	1	9
B	3	0	6	11
C	4	1	0	20
D	6	5	4	0

- (b) Apply divide and conquer strategy to perform quick sort on the following set of elements :

23, 9, 18, 45, 5, 16, 1, 17

[P. T. O.]

- (c) Define Huffman code. Generate optimal Huffman code for the following set of characters :

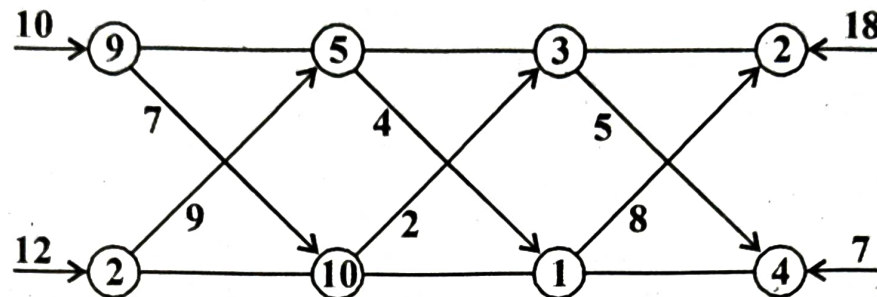
Characters	Frequency
B	9
A	5
D	13
C	12
F	20
E	16
H	30
G	25
I	40

5. (a) Given $X = \text{"acbaed"}$ $Y = \text{"abcadf"}$ $Z = \text{LCS}(X, Y)$. Find Z as longest common subsequence of X and Y using dynamic programming approach.

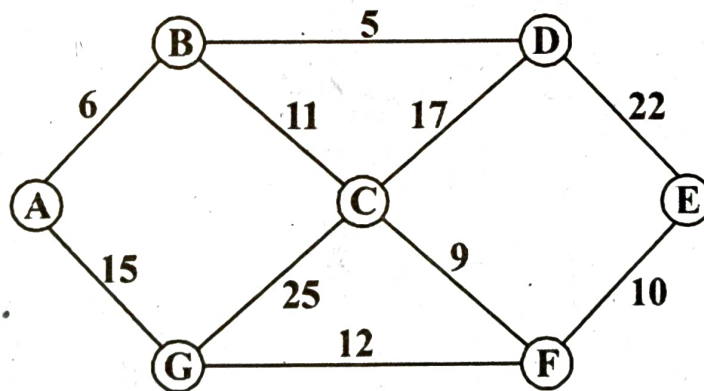
- (b) Explain sum of subset problem. Solve it for the following set using back tracking :

$$S = \{1, 2, 5, 6, 8\} \text{ and } x = 9.$$

- (c) Find the optimal solution for the following using assembly line scheduling :



6. (a) Explain assignment problem with the help of an example.
- (b) Find the minimum spanning tree for the following graph using kruskal's algorithm :



- (c) Explain how Bellman Ford algorithm is used for finding shortest path problem.
